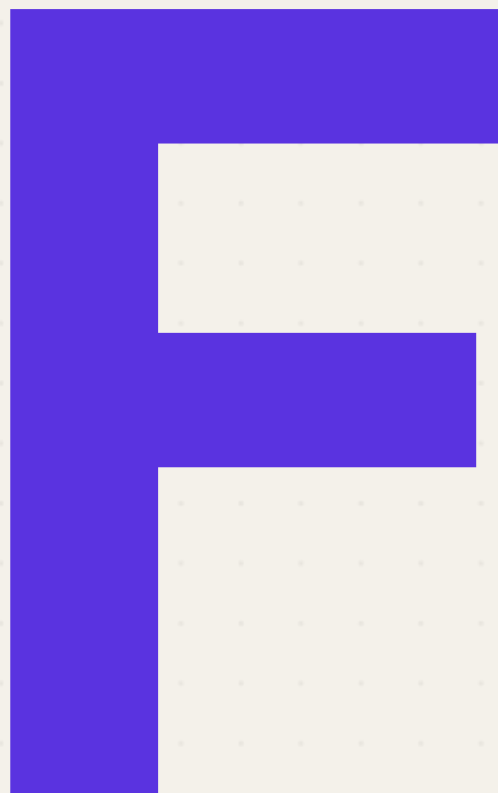




LV.0 → LV.1 · AI ENGINEER

# The 6 F's that move you from **Level 0** → **1** AI engineer.

Forward Propagation, Fine-tuning, Few-shot Prompting and more.  
**Plain-English** definitions, one swipe each, plus the tool the whole  
stack runs on.

FORWARD PROPAGATION • FOUNDATIONS

# Forward Propagation

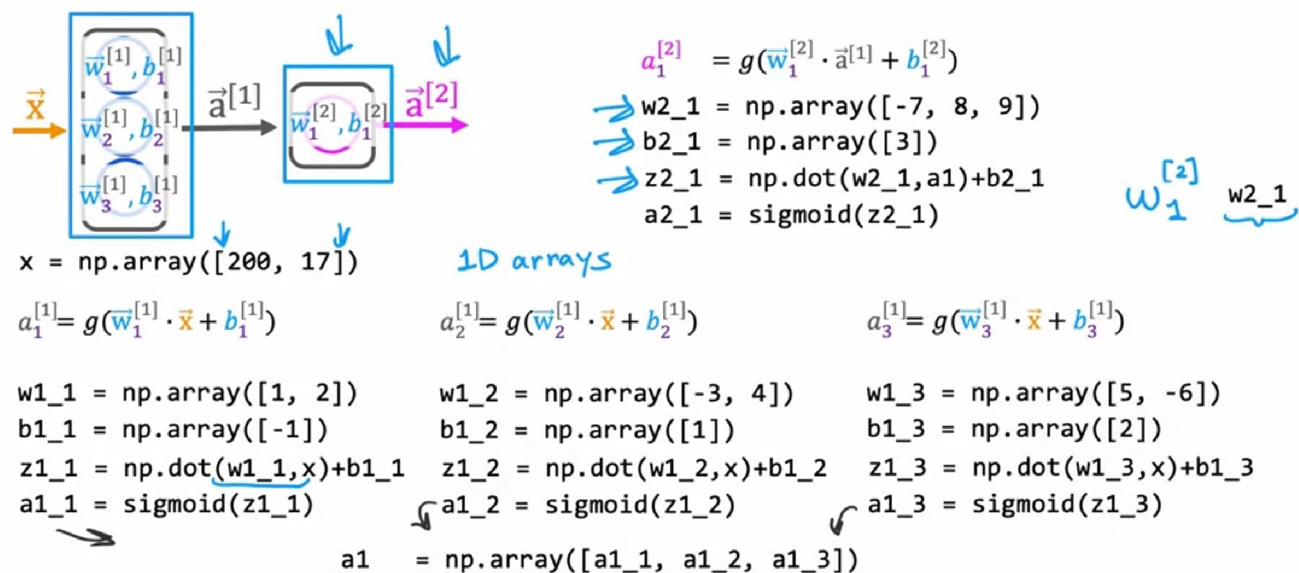
input • weights • prediction

This is the model thinking. Input flows forward through each layer, picking up weights, biases, and activations until a prediction drops out the far end. Backprop fixes mistakes later; **forward propagation is the guess**.

USED IN

- Neural networks
- Inference
- Deep learning

## forward prop (coffee roasting model)



INPUT → LAYERS → OUTPUT

FOURIER TRANSFORM • FOUNDATIONS

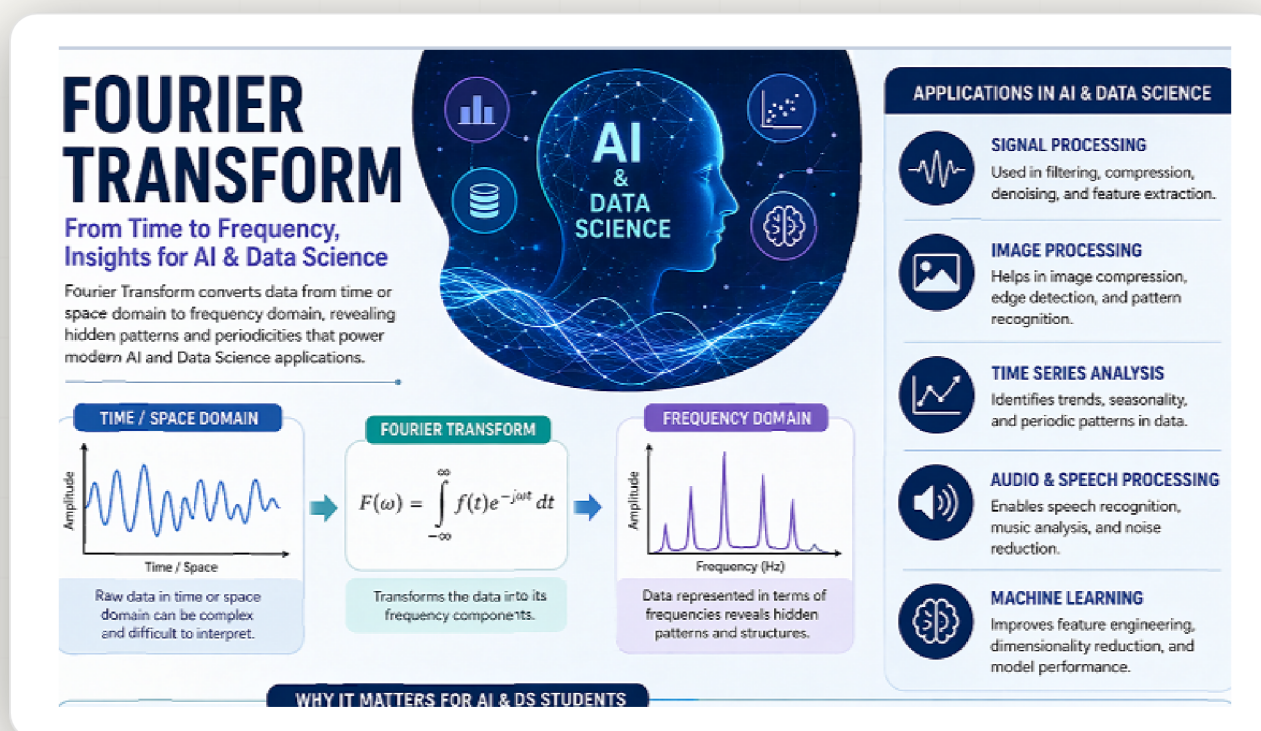
# Fourier Transform

time → frequency

Any signal, however messy, is really a stack of simple waves. The Fourier Transform **pulls it apart into frequencies**, which is how we compress images, clean audio, and engineer features for ML.

USED IN

- Signal processing
- Image compression
- Feature extraction



SIGNAL → SPECTRUM

FEW-SHOT PROMPTING • INFERENCE

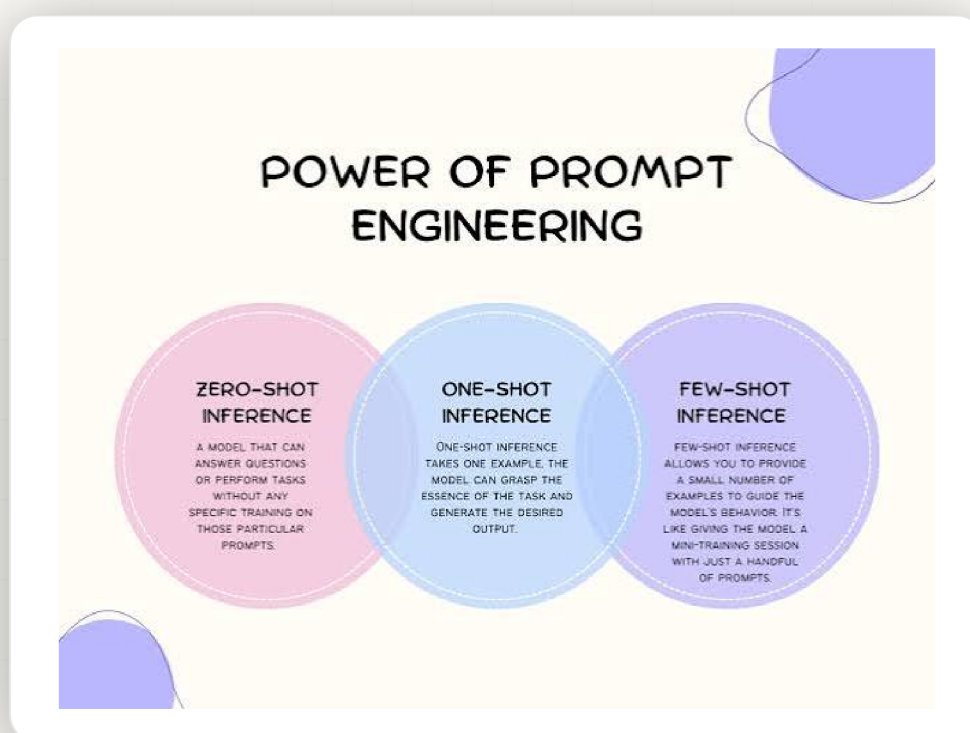
# Few-shot Prompting

show • examples • then ask

Want better answers? Show, don't tell. Drop **a few worked examples** into the prompt and the model copies the pattern, format, and tone, no training run required. It often beats zero-shot on niche tasks.

USED IN

- Prompt engineering
- In-context learning
- LLM apps



ZERO • ONE • FEW-SHOT

FINE-TUNING • LEARNING

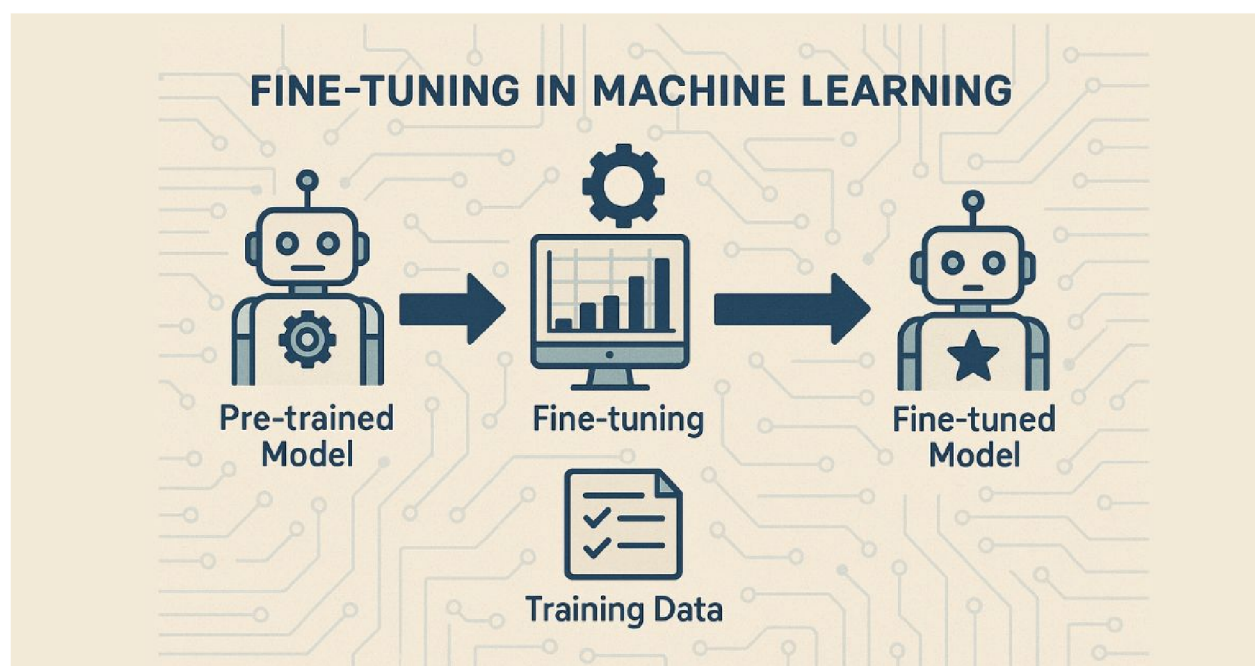
# Fine-tuning

pre-trained → specialized

Take a model that already knows a lot, then keep training it on your data. Fine-tuning **nudges a general model toward one domain or voice**, whether that is legal, medical, or your company's own tone.

USED IN

- Domain LLMs
- LoRA / PEFT
- Custom models



ADAPT TO A DOMAIN



FAITHFULNESS • EVALUATION

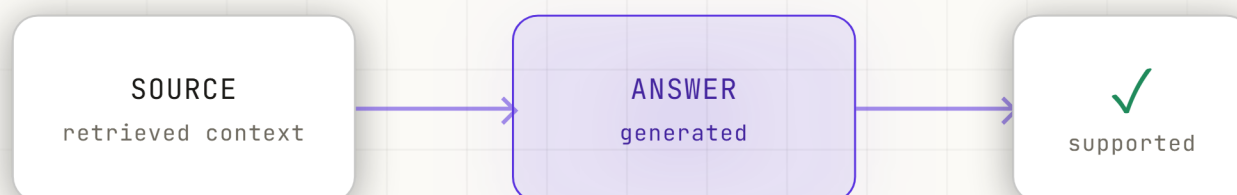
# Faithfulness

answer • backed by • source

Did the answer really come from the source, or did the model freelance? Faithfulness scores exactly that. In RAG and summarization, a high score means it **stayed honest to the context** it was handed.

USED IN

- RAG evaluation
- Hallucination checks
- RAGAS



GROUNDED CHECK

FLOATING POINT OPERATIONS • INFRASTRUCTURE

# FLOPs

how much • compute

Compute has a unit, and this is it. FLOPs count the **floating-point math a model burns** to train or run, setting the scale, speed, and price of systems like GPT and Claude. Much of the last decade was simply more of them.

USED IN

- Training cost
- GPUs / TPUs
- Scaling laws



↓ UP NEXT • TOOL

# Tool next.

Beyond the **6 F-words**, the one tool quietly moving the audio and video behind half the AI you use.

---

F



OPEN-SOURCE MEDIA FRAMEWORK

# FFmpeg

record • convert • stream • filter



If audio or video ever moved through software, FFmpeg probably touched it. This open-source Swiss Army knife **decodes, converts, and filters almost any format**, quietly powering players, pipelines, and AI data prep.

## WHAT IT IS

A universal audio/video framework and CLI.

## IN AI STACKS

Prepping audio/video for training & inference.

## WHY DEVS USE IT

One tool for any codec, format, or filter.

## PAIRS WITH

Whisper, OpenCV, PyTorch data loaders.

↓ UP NEXT • CHANNELS TO FOLLOW

# Spotlight next.

Two **high-signal, low-noise** channels worth a follow to keep your F-stack current.

---

F

HIGH-SIGNAL, LOW-NOISE

# Fireship

YouTube • @Fireship • by Jeff Delaney



Complex tech, explained before your coffee cools. Fireship turns GPT, agents, and new dev tools into **fast, funny 100-second videos**, the quickest way to stay current without the noise.

## WHAT THEY POST

Rapid-fire breakdowns of AI tools & trends.

## FIND THEM

YouTube @Fireship, X @fireship\_dev.

## WHY FOLLOW

Complex tech, explained in 100 seconds.

## BEST FOR

Staying current on AI & dev fast.

HIGH-SIGNAL, LOW-NOISE

# freeCodeCamp

YouTube • free, full-length courses



Millions of developers got their start here, for free. freeCodeCamp ships **project-based courses** and one of the largest coding channels anywhere, covering everything from web dev to AI.

## WHAT THEY POST

Free full-length AI & coding courses.

## FIND THEM

YouTube @freeCodeCamp, [freecodecamp.org](https://freecodecamp.org).

## WHY FOLLOW

Project-based learning at no cost.

## BEST FOR

Going deep on a topic, end to end.

# The **F** stack, in one card.

Save it for later. Drop a comment if there's an F-term you'd add.

01 **Forward Propagation** foundations

02 **Fourier Transform** foundations

03 **Few-shot Prompting** inference

04 **Fine-tuning** learning

05 **Faithfulness** evaluation

06 **FLOPs** infrastructure

+ **FFmpeg** tool

@ **Fireship** channel

@ **freeCodeCamp** channel



FOLLOW FOR THE FULL A TO Z

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**G Next**